OneEdge-Firmware Management App

Application Note

80654NT11889A Rev. 1 – 2022-06-14

APPLICABILITY TABLE

PRODUCTS	Platform Version ID
ME910C1-AU ME910C1-E1 ME910C1-E2 ME910C1-NA ME910C1-NV ME910C1-WW ML865C1-EA ML865C1-NA	30
ME910G1-W1 ME910G1-WW ME310G1-W1 ME310G1-WW ME310G1-W2 ML865G1-WW	37

CONTENTS

1. INTRODUCTION

1.1. Scope

This document describes how to upgrade the firmware of Telit Modules using FOTA functionality via OneEdge Application (FOTA APP)

1.2. Audience

This document is intended for those Users who need to upgrade the firmware of their modules taking advantage using the OneEdge services.

1.3. Contact Information, Support

For general contact, technical support services, technical questions and report of documentation errors contact Telit Technical Support at:

- *TS-ONEEDGE@telit.com*

Alternatively, use:

https://www.telit.com/support

For detailed information about where you can buy the Telit modules or for recommendations on accessories and components visit:

https://www.telit.com

Our aim is to make this guide as helpful as possible. Keep us informed of your comments and suggestions for improvements.

Telit appreciates the User feedback on our information.

1.4. Symbol Conventions

	Danger: This information MUST be followed, or catastrophic equipment failure or personal injury may occur.
	Warning: Alerts the user on important steps about the module integration.
	Note/Tip: Provides advice and suggestions that may be useful when integrating the module.
	Electro-static Discharge: Notifies the user to take proper grounding precautions before handling the product.

Table 1: Symbol Conventions

All dates are in ISO 8601 format, that is YYYY-MM-DD.

1.5. Related Documents

- ME910C1 AT Command Reference Guide: 80529ST10815A
- MExxx910C1 LwM2M AT commands User Reference Guide: 80529ST10974A
- MExxxG1 LwM2M AT commands Reference Guide: 80617ST11022A
- OneEDGE Getting Started: 1VV0301585

2. OVERVIEW

The aim of this document is to describe the Firmware update of a Telit OneEdge modem using the FOTA Application. Firmware Over-the-Air (FOTA) allows Users to wirelessly update the module operating firmware.

The main feature of the Firmware Management App is the possibility to perform the upgrade without any on site intervention. The firmware upgrade process can be applied remotely on multiple devices. OMA Lightweight M2M (LwM2M) provides a specific object called "Firmware Update" to instruct devices to download and apply firmware updates through FOTA.

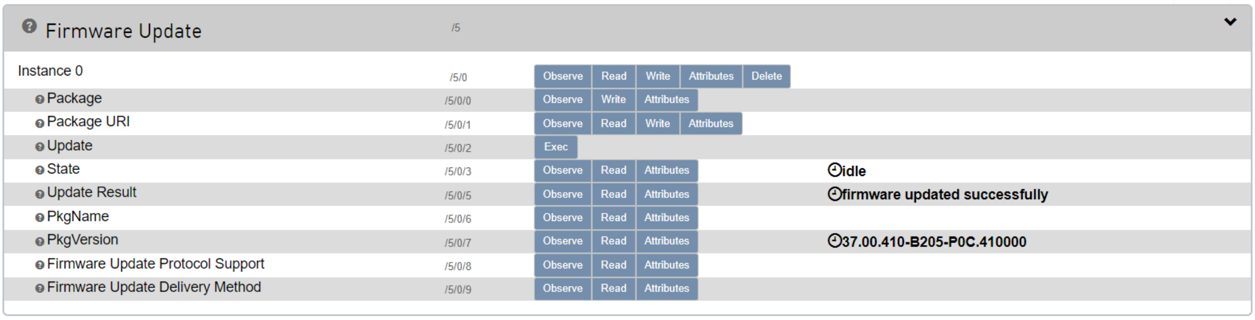


Figure 1 : OMA Object n°5

Telit Firmware Management Application works in accordance with this object and the OMA state machine, to handle the whole firmware updating process.

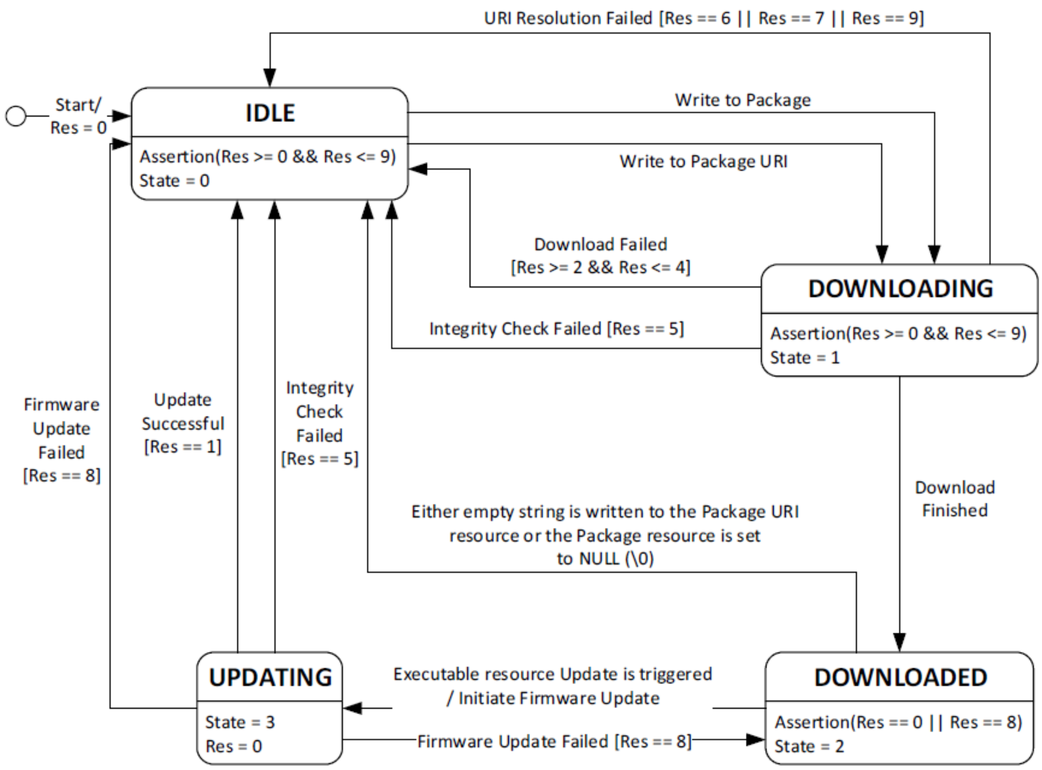


Figure 2: OMA FOTA State Machine

3. HOW TO UPGRADE THE FIRMWARE OF ONEEDGE DEVICE

3.1. Log in to Firmware Management App

The App Center allows the User to access the Apps enabled in their profile. To start using the App Center, open your web browser and visit https://appcenter.telit.com/login

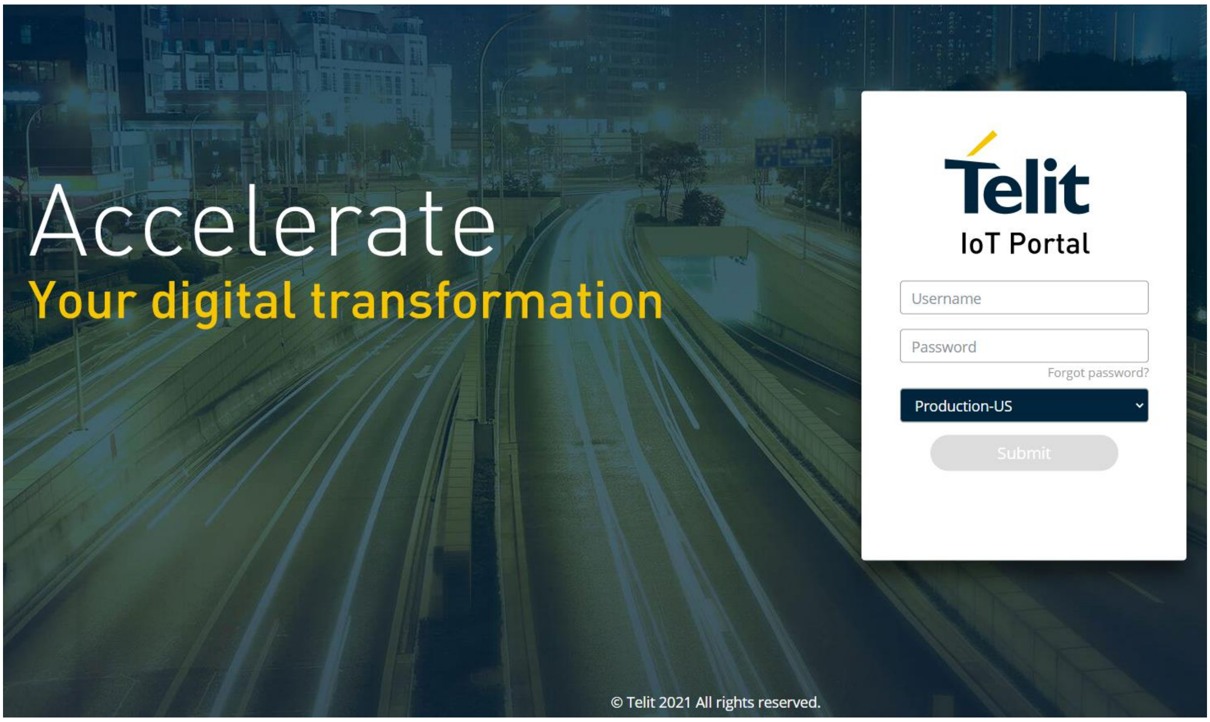
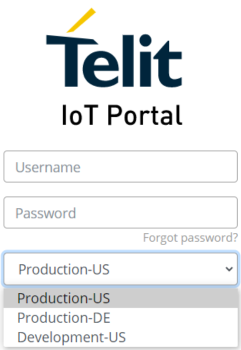


Figure 3: Access Page

Warning: Make sure to select the right portal



After login, the enabled Apps are shown in the homepage. Each App within the App Center provides a specific service.

The Firmware Management Application is one of them.

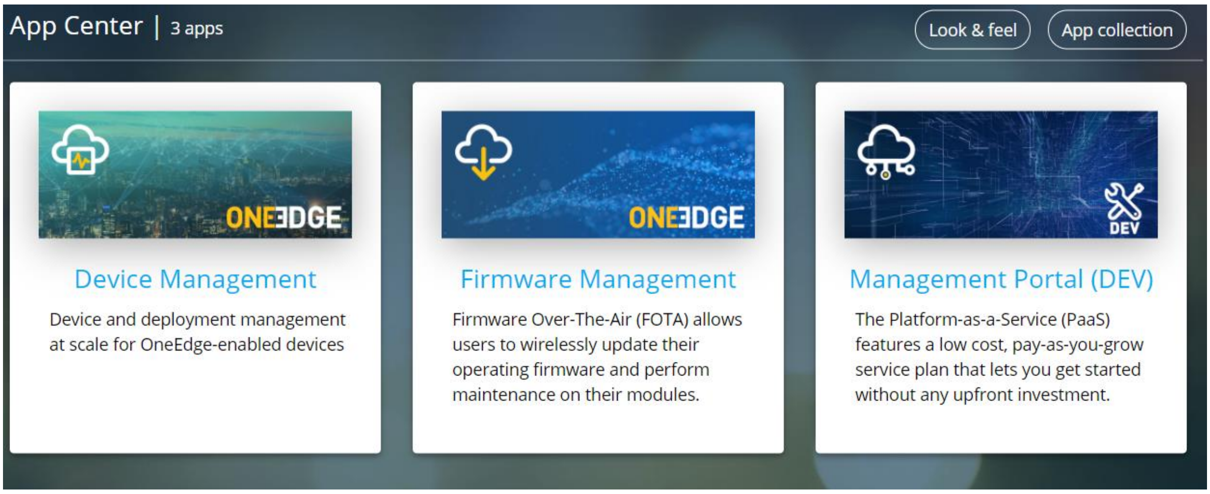


Figure 4: App Center

3.2. Firmware Management App Home

The Home page is organized in four different sections, each section contains specific information (Figure 4.3), as described below:

- 1 DEVICES all the things present in the customer organization, split by thing definition. Each model is displayed with its Firmware version
- 2 RECENT CAMPAIGNS list of the recent campaigns
- 3 CAMPAIGN OVERVIEW displays the status of the campaigns configured in the system
- 4 RECENT ACTIVITY log of the recent activities in the FOTA Campaign section.

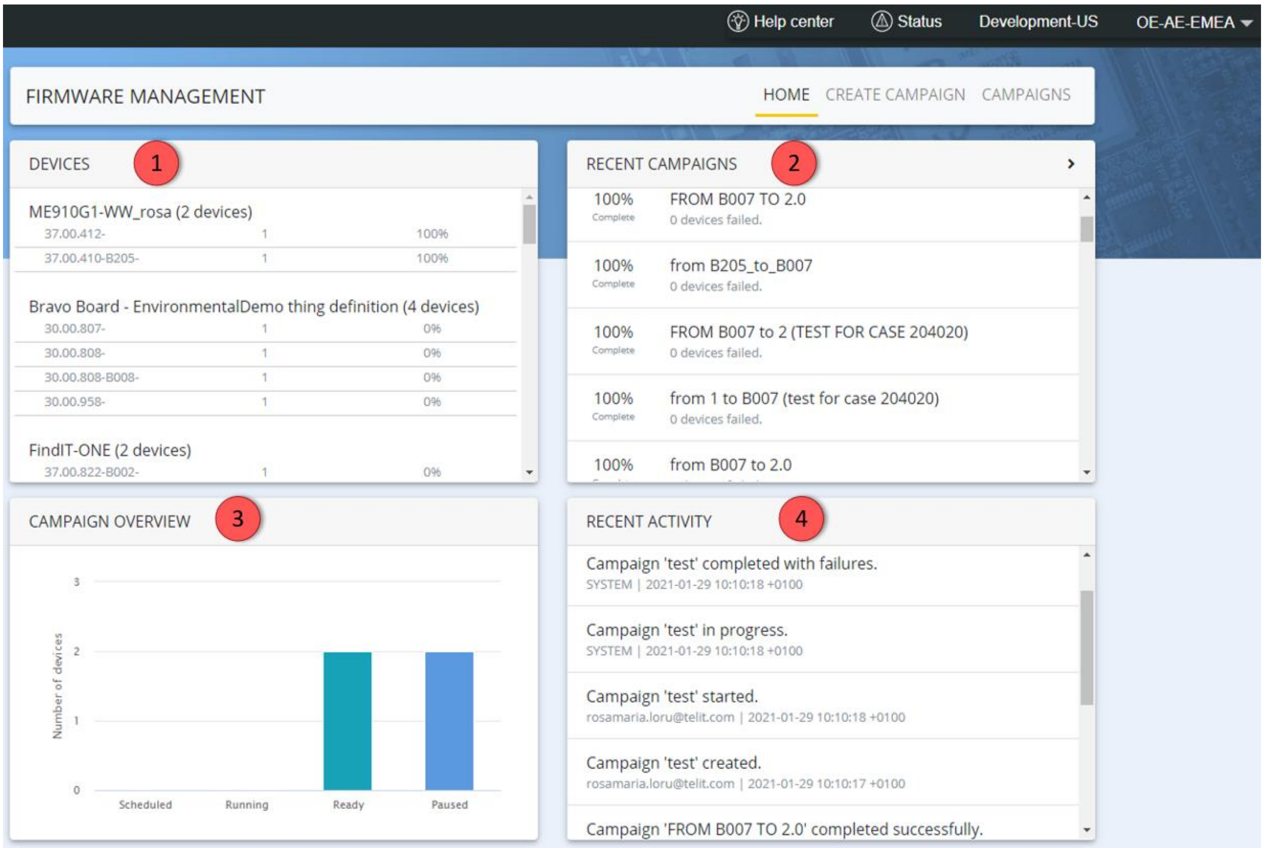


Figure 5: Home

3.3. Creating a FOTA Campaign

The FOTA campaign allows to apply the same firmware update to a set of devices that belong to a specific device model.

To create a FOTA campaign, select the **CREATE CAMPAIGN** tab and follow the four steps.

3.3.1. Step 1: Model selection

The User should select the specific model name (thing definition) from the drop-down menu that lists all the thing definitions present in the customer organization.

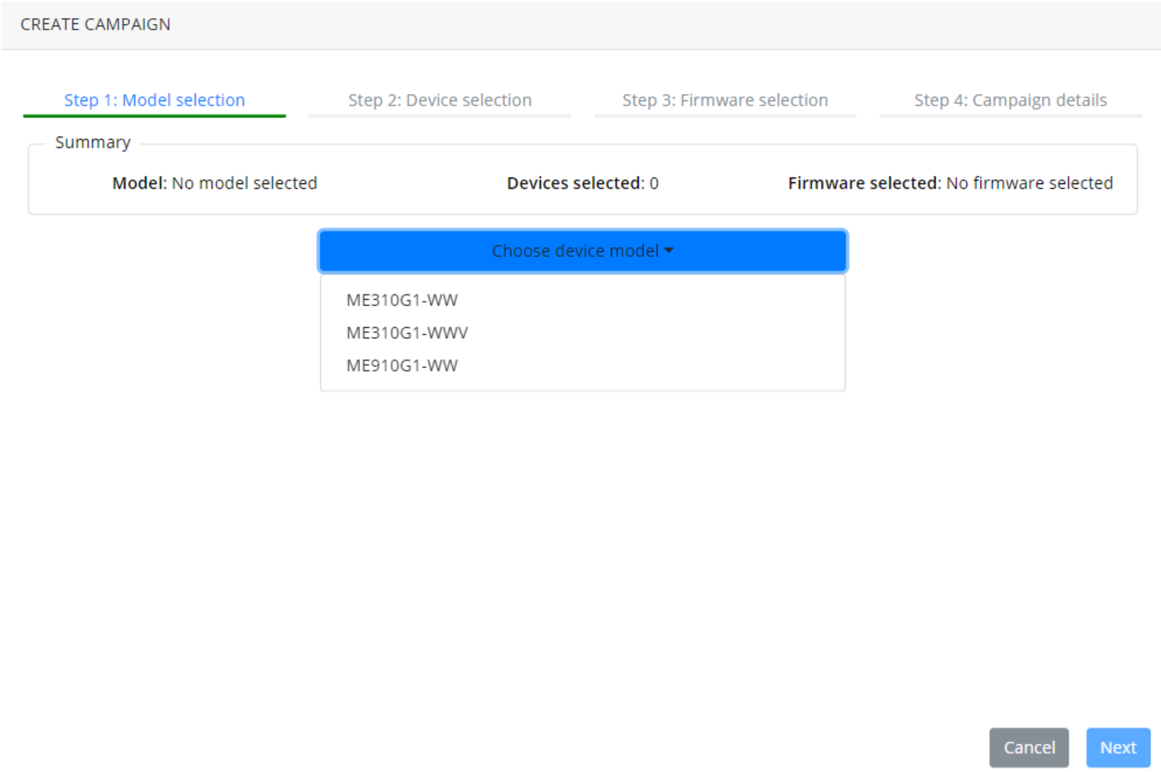


Figure 6: Device Model Menu

Once the model is selected, press **Next** button to procede.

3.3.2. Step 2: Device selection

This section allows to select the devices that will be included in this firmware upgrade campaign.

Two different options can be selected:

- 1 Device list
- 2 Upload devices

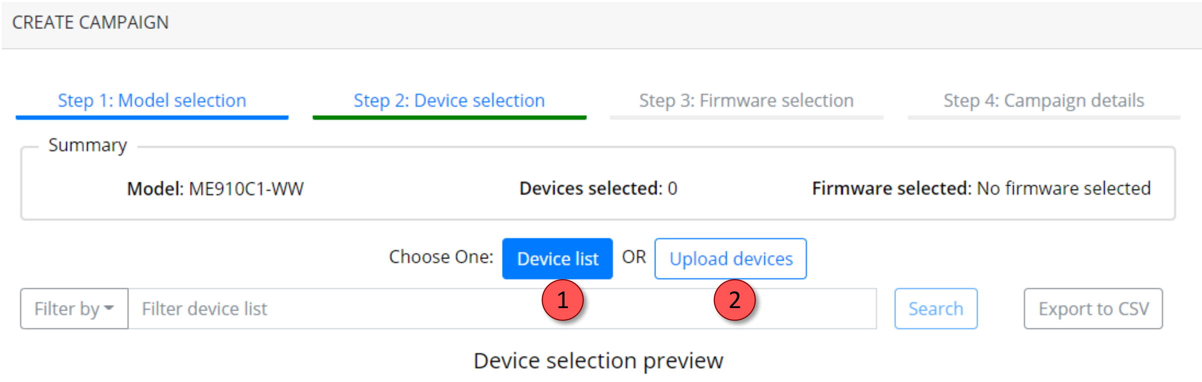


Figure 7: "Filter by" Menu

3.3.2.1. Device List

The **Device list** allows to filter the devices in three different ways:

- string (for example device name or IMEI), choosing the **Query** option
- by firmware version, choosing **Current firmware version** option
- or by tags, selecting **Tags** and entering the name of tag or tags

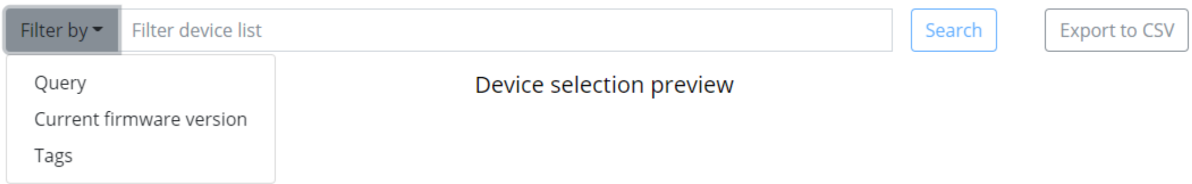


Figure 8: "Filter by" Menu

All these criteria will be activated by pressing the **Search** button and the filtered devices will be listed in the device selection preview.

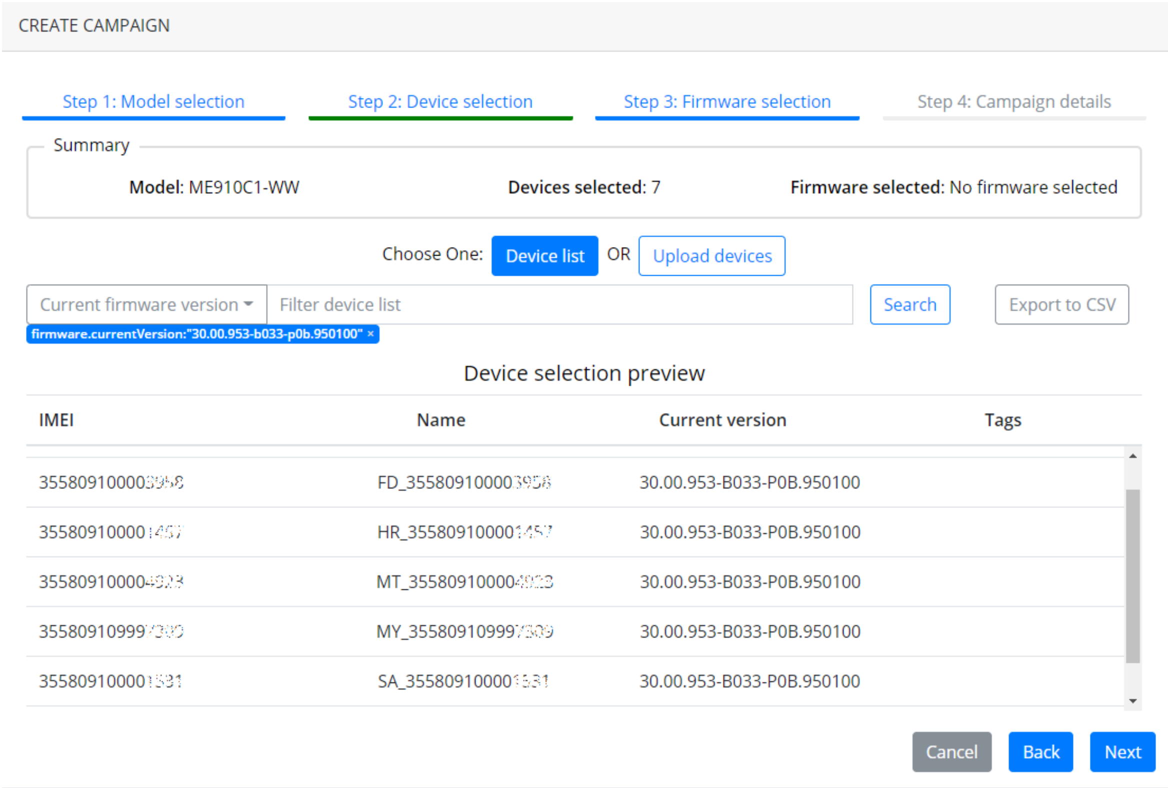


Figure 9: List of Devices - Example

Warning: Please be aware that only the onboarded things appear in this list. The User must ensure proper onboarding before starting.

3.3.2.2. Upload Devices

This feature allows to edit offline the list of devices to include in the campaign and import it via CSV file.

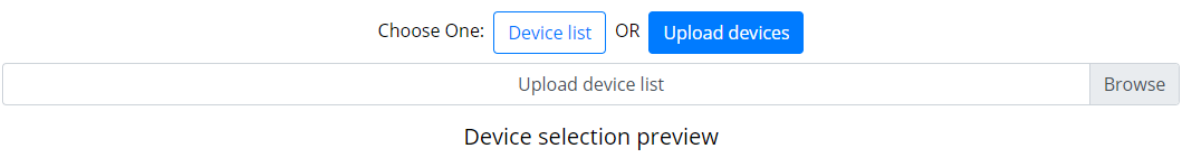


Figure 10 : Upload CSV File

It is very usefull in case of the User already has the formatted list of things or the selection throught "Filter by" fails to retrieve the correct results.

3.3.3. Step 3: Firmware Selection

The Firmware versions available in this menu are those associated to the device model.

Those are the Firmware version that can be applied to the devices.

Note: The Firmware version is associated to a delta file. It is a binary package which contains only the differences between two firmware images: the delta size is smaller than the entire image.

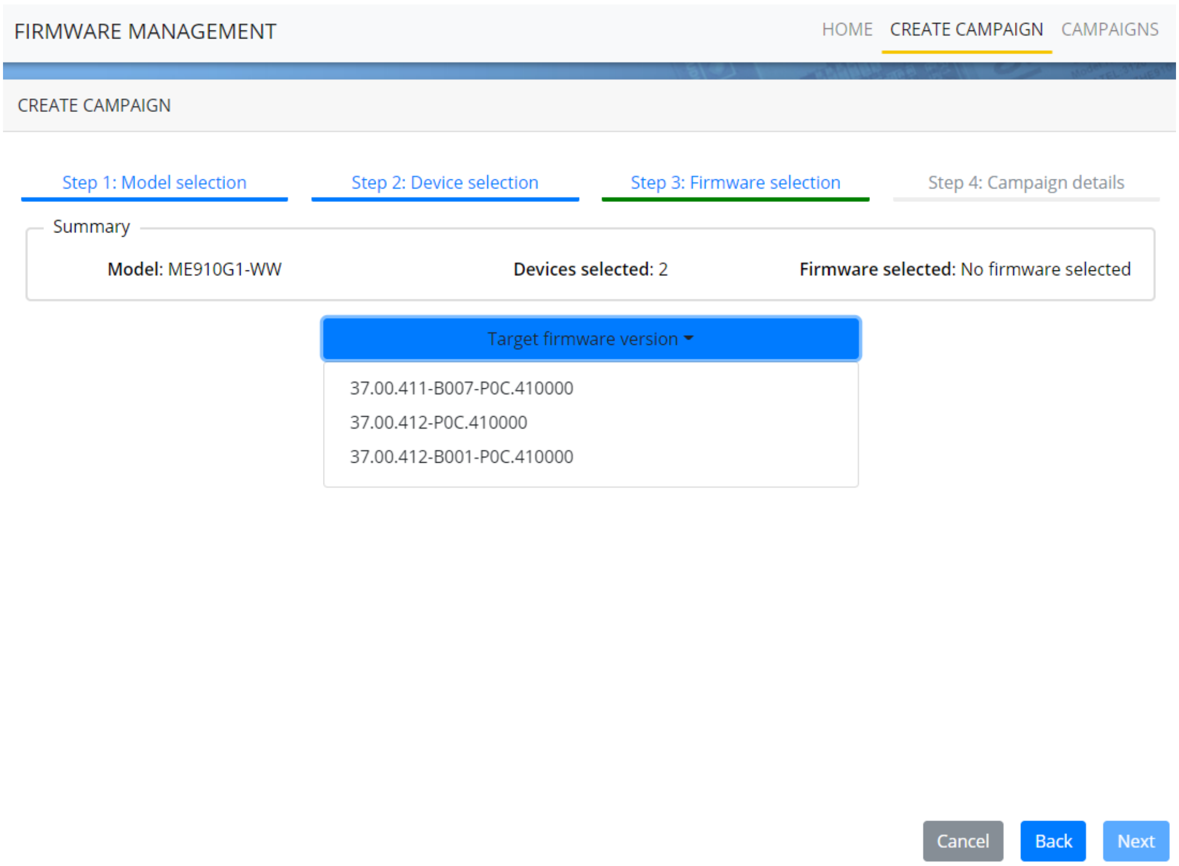


Figure 11 : Firmware Selection

Warning: This drop-down menu may be blank or other firmware versions may be required. The User should request them from OneEdge technical support team (ts-oneedge@telit.com) before starting the Firmware upgrade process.

After selecting the Firmware Version click Next to proceed to scheduling the campaign.

3.3.4. Step 4: Campaign Details

This section is the last one and allows to start immediately the firmware upgrade on the selected devices or schedule it for later.

Warning: To enable the **Create campaign** button, the user must agree terms and conditions:

Addtional configuration parameters are described below:

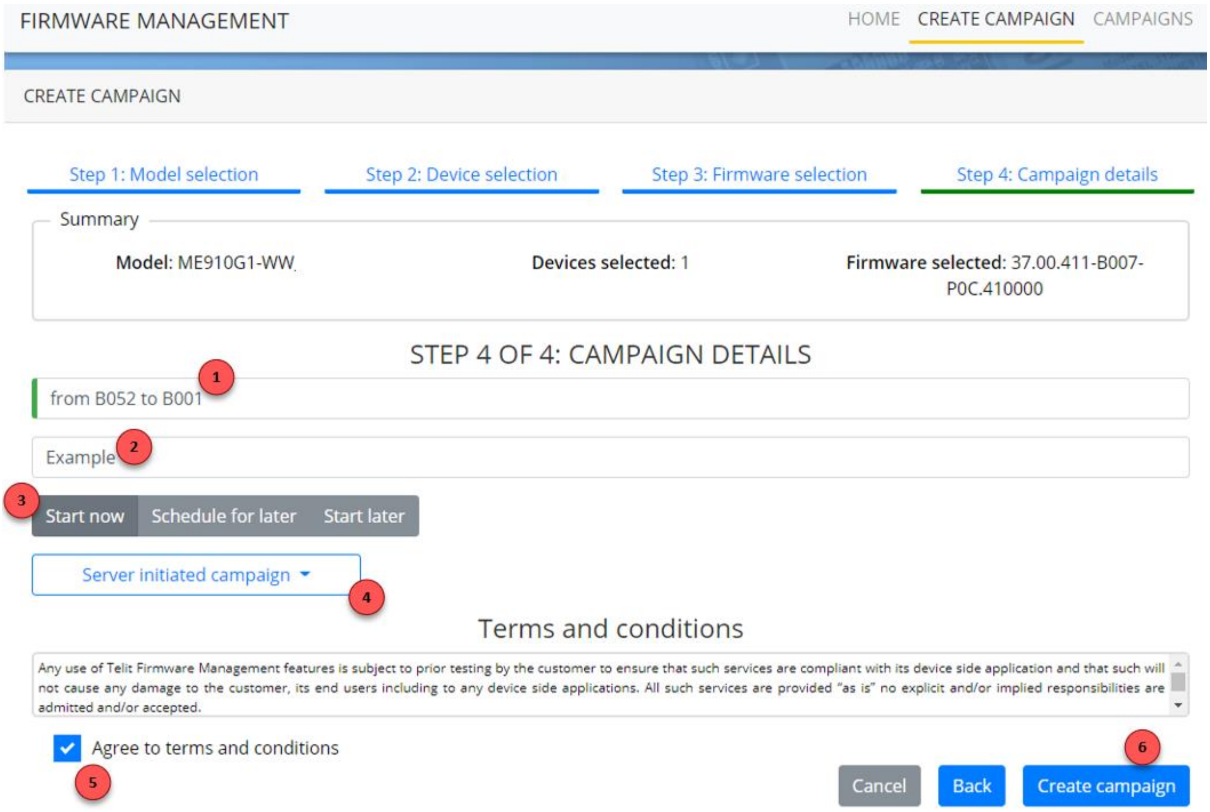


Figure 12: Firmware Selection

- 1 **NAME**: Enter the name of the campaign;
- 2 **DESCRIPTION:** (optional) a brief description of the campaign;
- 3 **START NOW/SCHEDULE FOR LATER/START LATER**:
  - start the campaign immediately when it's created;
  - schedule the campaign. Selecting this option, the User will indicate the time/date through the following tabs

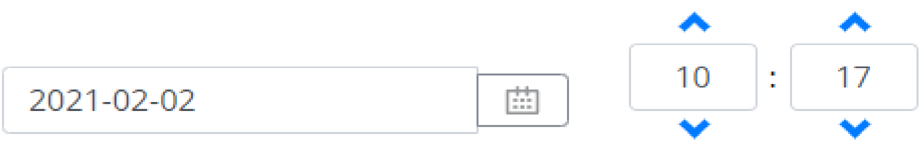


Figure 13: Time Window Selection

  - create and save the campaign for later. The campaign can be started by selecting it from list in the "Campaigns" tab.

- 4 **SERVER INITIATED CAMPAIGN**: (optional) a wakeup SMS is sent to the device to immediately start the upgrade. Otherwise, the upgrade process starts only when the device connects to the platform (e.g. when publishing data or updating the registration);
- 5 check box to agree to terms and conditions;
- 6 **CREATE CAMPAIGN**: start the campaign or save for later.

At the end of the Campaign, the devices will boot up running the target Firmware Version

3.4. Campaigns

The campaigns view provides overall progress of all campaigns in your organization, along with the current status and the date each campaign is scheduled or started.

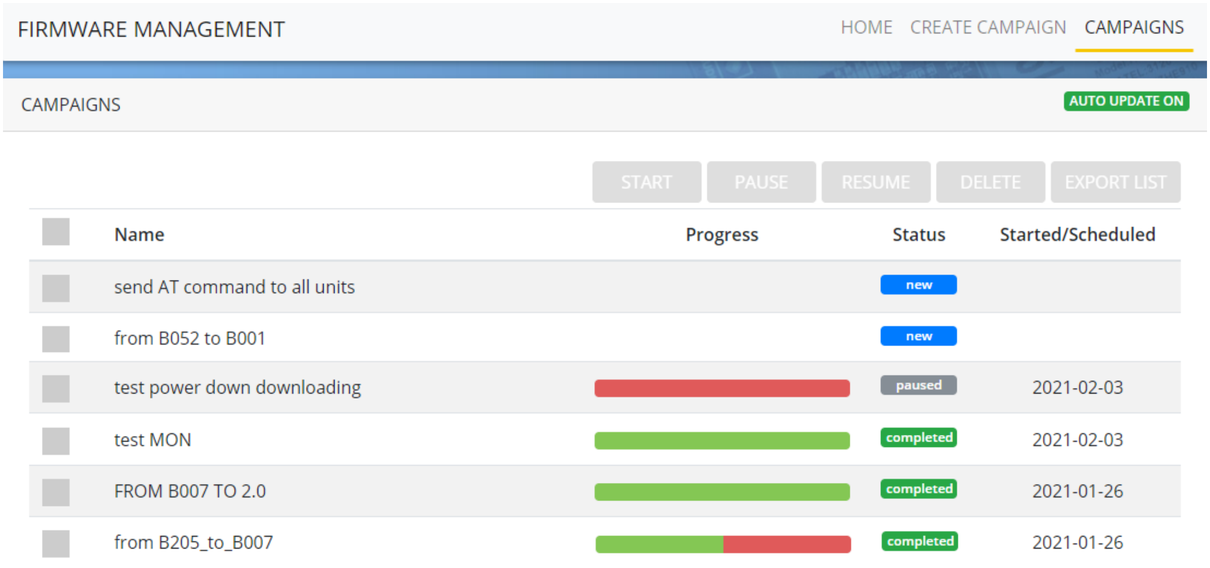


Figure 14: Campaigns History

By selecting one or more campaigns, the User will be able to perform some actions (pause/resume, delete, start campaign or exporting campaigns list to a CSV file).

More details of a campaign are available by opening a specific one, as in the picture below:

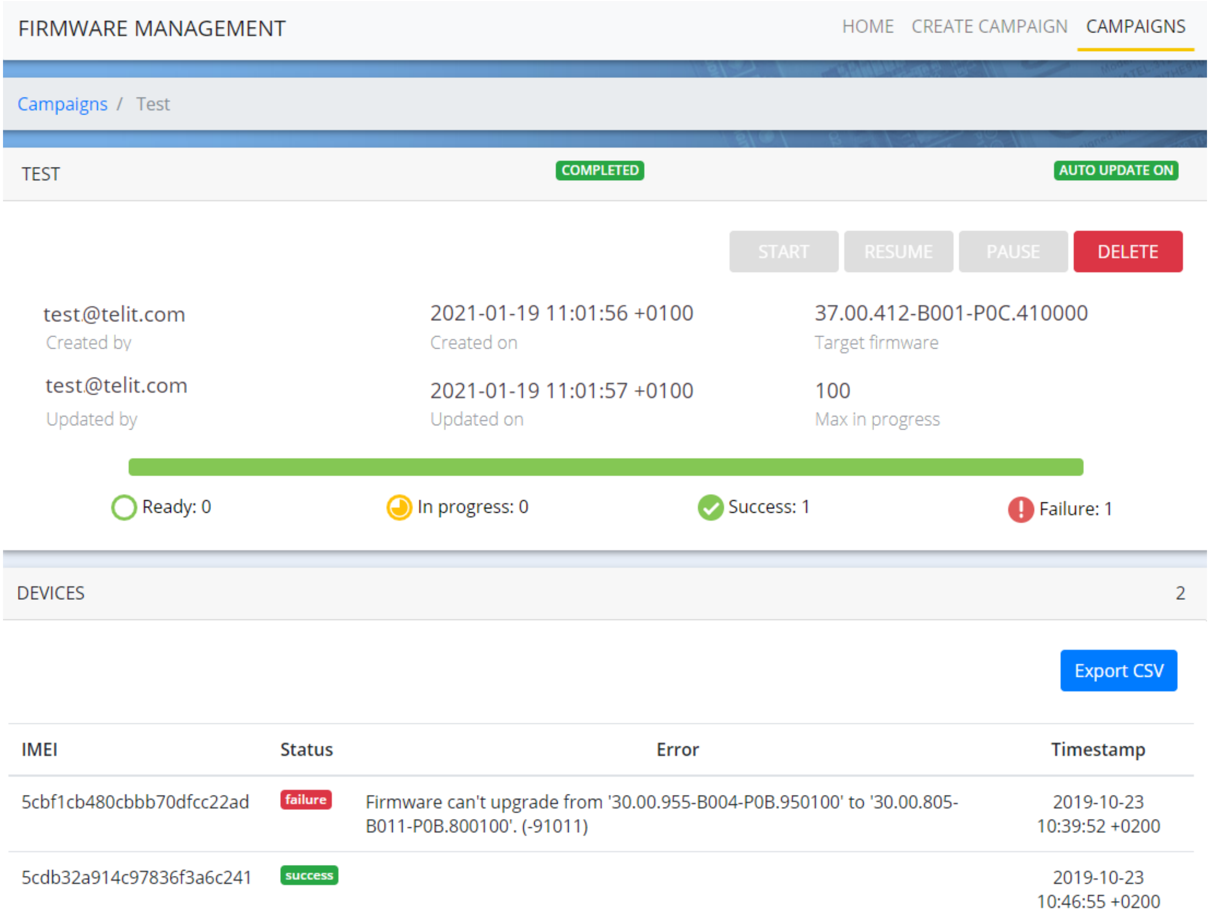


Figure 15: Campaign Details

Note: please be aware that each Campaign has a timeout of 24 hours from the moment it starts.

4. FOTA PROCEDURE DETAILS

This section describes how the Firmware Management App server interacts with the LwM2M client to manage all the steps required to successfully complete the firmware upgrade process.

The FOTA state machine was developed according to OMA LwM2M specifications.

There are three main cases:

- Normal mode (no ACK): the LwM2M client receives the FOTA upgrade request without any possibility to accept or reject the process.
    - the host application accepts the disruption caused by the upgrade process;
    - Available for all firmware versions.
- ACK mode: the host can accept the FOTA upgrade through the ACK messages
    - The host application controls the process step by step by providing an ACK.
    - Available for modules with firmware versions starting from 37.00.XX1 (for MExxxG1 devices), 30.00.XX7 (for MExxxC1 devices).
- Reject mode: the host can reject the FOTA upgrade (FOTA reject)
    - The host application rejects the FOTA upgrade if it cannot be interrupted, for example when it is sending data or performing time-consuming tasks;
    - Available for modules with firmware versions starting from 37.00.XX3 (for MExxxG1 devices), 30.00.XX9 (for MExxxC1 devices).

To perform the FOTA process, the LwM2M client needs to be registered on the LwM2M server and when the process is complete, the LwM2M client will bootstrap again automatically.

The entire duration of the process depends on the delta file size, signal quality, network congestion and bitrate (typically ranging from few minutes up to 15-20 minutes).

During the update, the modem may not be fully operational and the serial port could be unavailable, so it is strongly recommended to wait for the end of the process.

Warning: Please be aware that, after the new software version is applied, the LwM2M client perform a new bootstrap procedure and this might cause loss of stored data or settings.

4.1. Normal Mode: FOTA with "No ACK"

Configuration of the LwM2M client in "Normal mode" using the command:

```
AT#LWM2MFOTACFG=0,0
OK
```

which is the default setting, the host does not have the possibility to interact with the FOTA process once the User starts the campaign.

When the module connects to the server, the download of the delta file is triggered immediately and the campaign changes to "In progress" state.

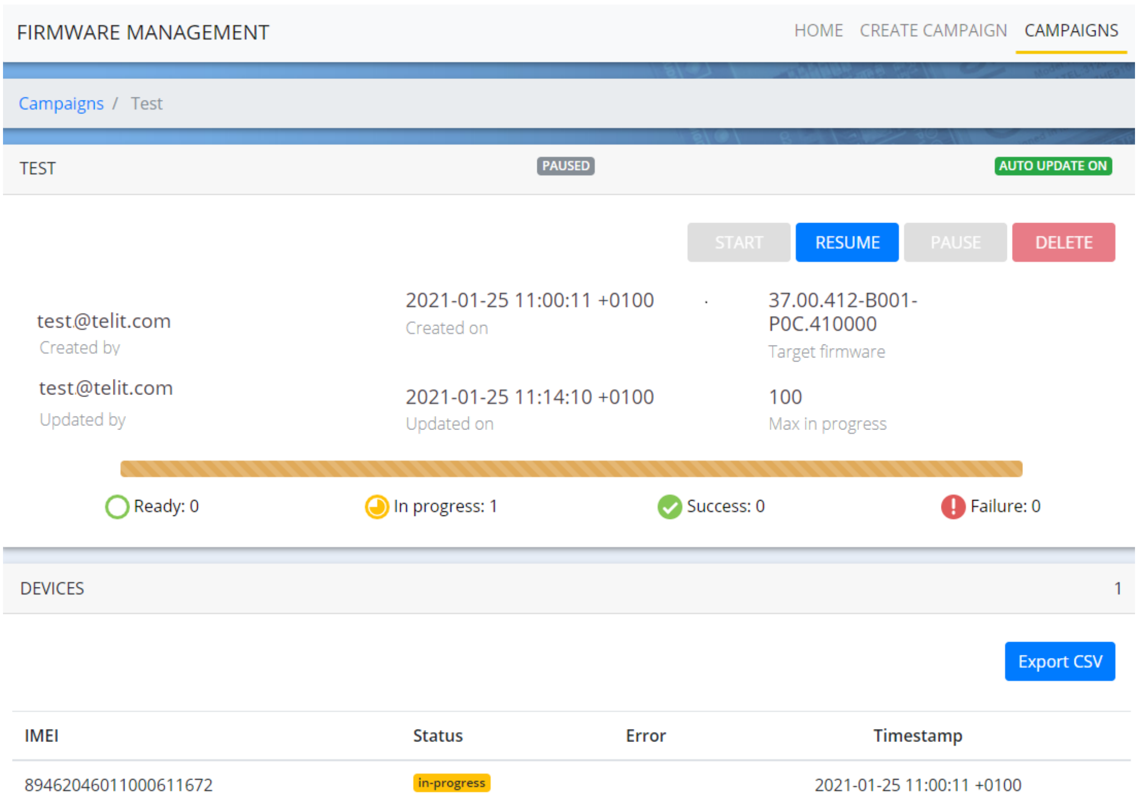


Figure 16: FOTA Campaign in Progress

The module enters the upgrade process without the possibility to interrupt it.

The host application can detect the start of the FOTA process by activating the monitoring on the object 5 “Firmware Update”.

The host activates the monitoring sending:

```
AT#LWM2MMON=1,5
OK
```

When the User starts the FOTA campaign on the FOTA app, the host receives the following URC message:

```
#LWM2MMON: UPD,"/5/0/1/0"
```

From now on, the host is aware that the module is downloading and applying the new firmware image until this message will be received:

```
#OTAEV: Module Upgraded To New Fw
```

Remember that also the serial port cannot be used until the above URC is sent.

The entire process ends when the module automatically reconnects to the LwM2M server after some power cycles and the registration process.

When the LwM2M client is connected to the server again, the FOTA campaign ends on the FOTA App and shows the status “Completed”.

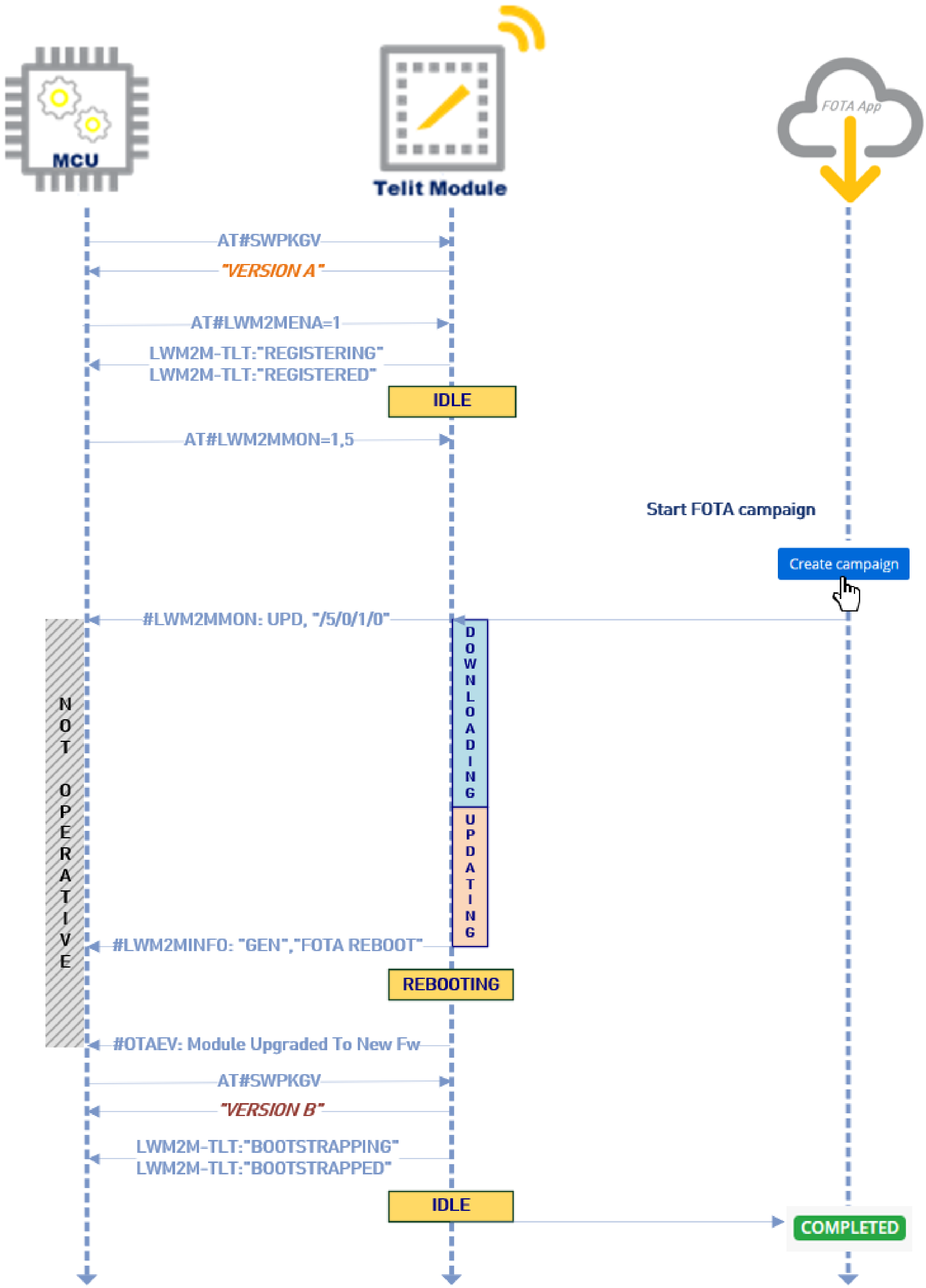


Figure 17: No ACK Mode

4.2. ACK Mode

The host can configure the LwM2M client to confirm FOTA operations step by step.

The following AT command is used to configure the FOTA ACK mode:

```
AT#LWM2MFOTACFG=0,3
OK
```

When the User starts the FOTA campaign on the FOTA app and the process is triggered on the module, the host receives this URC:

```
#LWM2MFOTARING: 0,"DOWNLOAD"
```

At that point the FOTA state machine waits for an ACK before downloading the delta.

Warning: All these operations could be performed or not in real time, depending of the thing settings (in particular the lifetime of the Device profile applied), as described in chapter "6.1. Timing conditions"

The host can confirm the operation immediately or acknowledge it later through the following command:

```
AT#LWM2MFOTAACK=0,1
OK.
```

Then the FOTA state machine starts downloading the delta image. After some time, when the download is complete, the host receives the following URC:

```
#LWM2MFOTARING: 0,"UPDATE"
```

Now the state machine waits for confirmation from the host before applying the new image.

As in the previous step, the host can confirm or delay the updating process immediately.

The command to confirm the operation is the same:

```
AT#LWM2MFOTAACK=0,1
OK
```

The FOTA machine starts to apply the delta image and the host receives, after a few seconds, the reboot confirmation from the LwM2M client:

```
#LWM2MINFO: "GEN","FOTA REBOOT"
```

After some reboots, the process ends and the host receives these URCs:

```
#OTAEV: Module Upgraded To New Fw
LWM2M-TLT:"BOOTSTRAPPING",SSID=0,"coaps://bs.telit.io"
LWM2M-TLT:"BOOTSTRAPPED",SSID=0,"coaps://bs.telit.io"
```

Finally, the LwM2M client registers again on the OneEdge portal and the FOTA campaign shows the status “Completed”.

Note: This is the most commonly used mode because it allows the host to control the impact and disruption on the service.

The figure below shows the entire process:

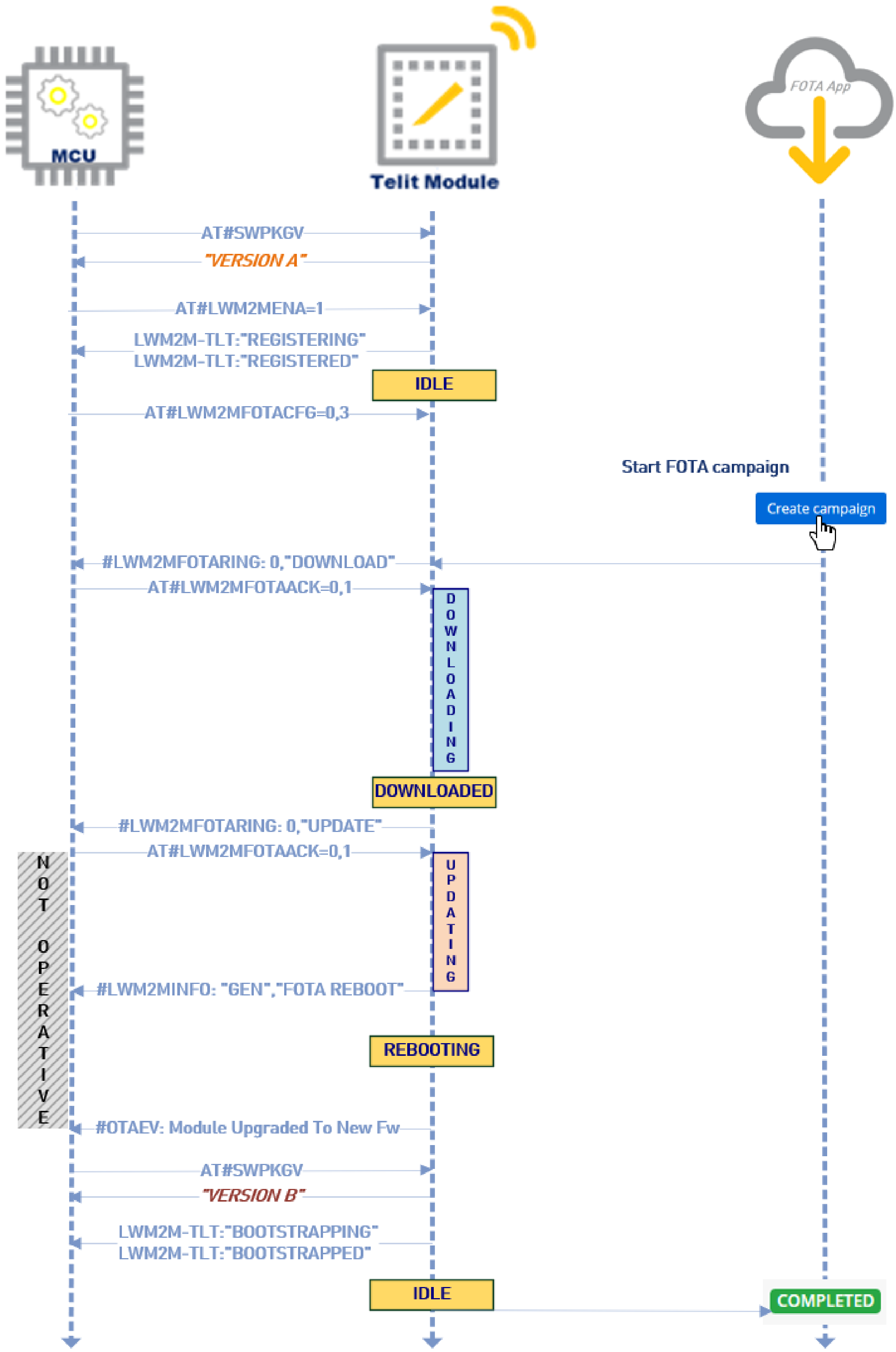

MCU
Telit Module
FOTA App
AT#SWPKGV
"VERSION A"
AT#LWM2MENA=1
LWM2M-TLT:"REGISTERING"
LWM2M-TLT:"REGISTERED"
IDLE
AT#LWM2MFOTACFG=0,3
Start FOTA campaign
Create campaign
#LWM2MFOTARING: 0,"DOWNLOAD"
AT#LWM2MFOTAACK=0,1
DOWNLOADING
DOWNLOADED
#LWM2MFOTARING: 0,"UPDATE"
AT#LWM2MFOTAACK=0,1
NOT OPERATIVE
UPDATING
#LWM2MINFO: "GEN","FOTA REBOOT"
REBOOTING
#OTAEV: Module Upgraded To New Fw
AT#SWPKGV
"VERSION B"
LWM2M-TLT:"BOOTSTRAPPING"
LWM2M-TLT:"BOOTSTRAPPED"
IDLE
COMPLETED

Figure 18: ACK Mode

4.3. FOTA Rejected

Starting with Software Versions 37.00.XX3 (for MExxxG1 devices) and 30.00.XX9 (for MExxxC1 devices), the possibility for the host to reject the FOTA process is available.

In this way the host can configure the LwM2M client to automatically reject the FOTA upgrade request when it does not want to manage the upgrade.

Warning: please note that the "Reject mode" requires to configure a new FOTA Campaign after its failure. On the contrary, the "ACK Mode" allows to use the same campaign simply by delaying the `AT#LWM2MFOTAACK=0,1`

The AT command to use is:

```
AT#LWM2MFOTACFG=0,4
OK
```

When the FOTA process is triggered on the module, the host receives the following URC message:

```
#LWM2MFOTARING: 0,"FOTA REJECTED"
```

and the campaign finishes reporting "failure" status, as showed in the figure below:

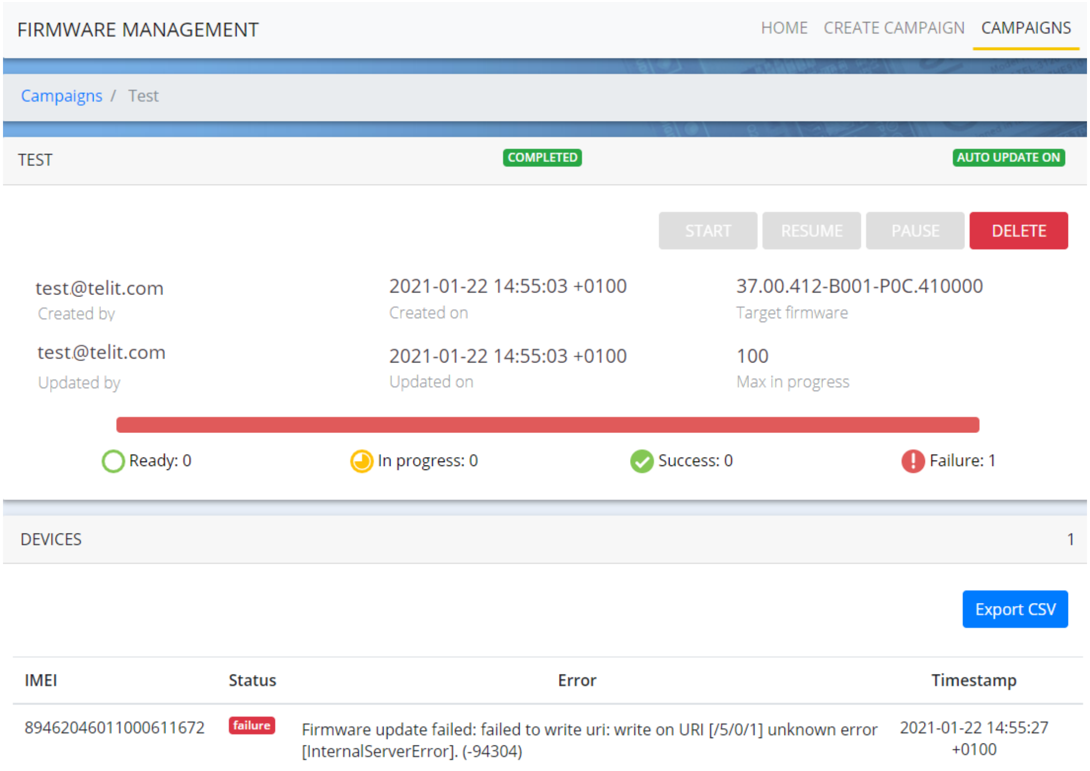


Figure 19: FOTA App – Rejected

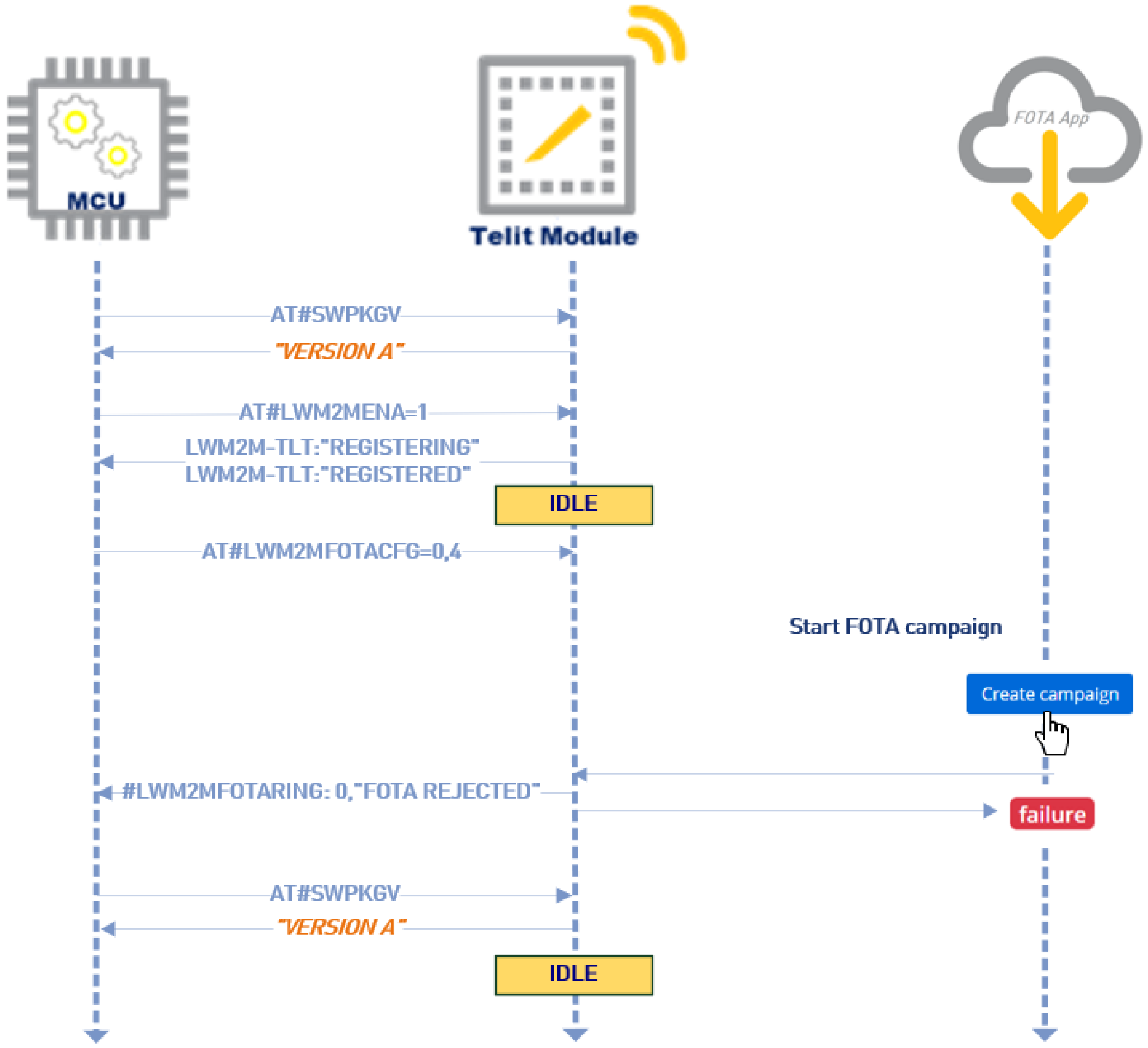


Figure 20: FOTA Reject

4.4. AppZone

In all the 3f lows described above the role of the MCU can be replaced by an application running in the AppZone environment.

For more information contact the TS EMEA support (ts-emea@telit.com).

5. FINAL CONSIDERATIONS AND CAVEATS

5.1. Timing Conditions

Normally the delta file download may take an amount of time which depends on multiple factors such as file size, network technology and operator, signal strength and others.

Furthermore, the delta file flashing could take up to fifteen minutes to be completed.

Please consider that this expected time window also depends on the lifetime of the LwM2M client and could be increased according to this rule:

Maximum expected timeout= LwM2M lifetime + download time + delta image flashing

Note that campaigns have a 24 hour timeout from the moment they start.

5.2. Stored Data

After a FOTA update, the host application does not need to reload the .xml files containing the definitions of the custom defined objects. The LwM2M client maintains all these definitions but the data values stored in the various objects/resources could be lost during the firmware upgrade operation.

Also instances and resource data may be deleted, so the host should save the relevant custom objects data before initiating the FOTA process and restore them after the upgrade.

6. FAIL CONDITIONS

In the event of an unexpected power failure, for example due to a too low battery level or mains failure, the FOTA process can be recovered by the LwM2M client or by the host, according to the following procedures.

6.1. Power Down During Delta Download

If a failure occurs during the delta downloading, the FOTA campaign fails and the User must restart it from scratch because the delta image is not stored locally.

The first piece of information that the host must retrieve from the LwM2M client is the actual firmware version, obtained through the command:

```
AT#SWPKGV
37.00.412-P0C.410000
```

called “Version A” for simplicity.

To evaluate the status of the LwM2M client and to understand how long it remains down, the host must perform a check to verify when the LwM2M client goes up again and also the status of the FOTA process.

The host should wait for the URC:

- `#LWM2MFOTARING: 0,"UPDATE"` in case of ACK mode, or
- `#LWM2MINFO: "GEN","FOTA REBOOT"` in normal mode.

to detect if the delta downloading is finished.

If the host does not receive one of these URCs within a timeout of approximately 15 minutes, it must check the LwM2M client status with the command:

```
AT#LWM2MENA?
```

So, if no response is provided, it could mean that the modem is still powered off.

The host may re-issue this command at regular intervals of time , waiting for the modem to power on and restart the LwM2M client again (for example the battery is replaced with a fully charged one). When the modem is switched on again, the LwM2M client replies with:

```
#LWM2MENA: 0
```

Now the host can check the current firmware version to evaluate the real state of the FOTA process, with the command:

```
AT#SWPKGV
```

If the retrieved software version is the same as before (Version A), it means that the FOTA process failed.

When the LwM2M client is re-registered on the portal, the FOTA operation on the device is reported as failed, as in figure below:

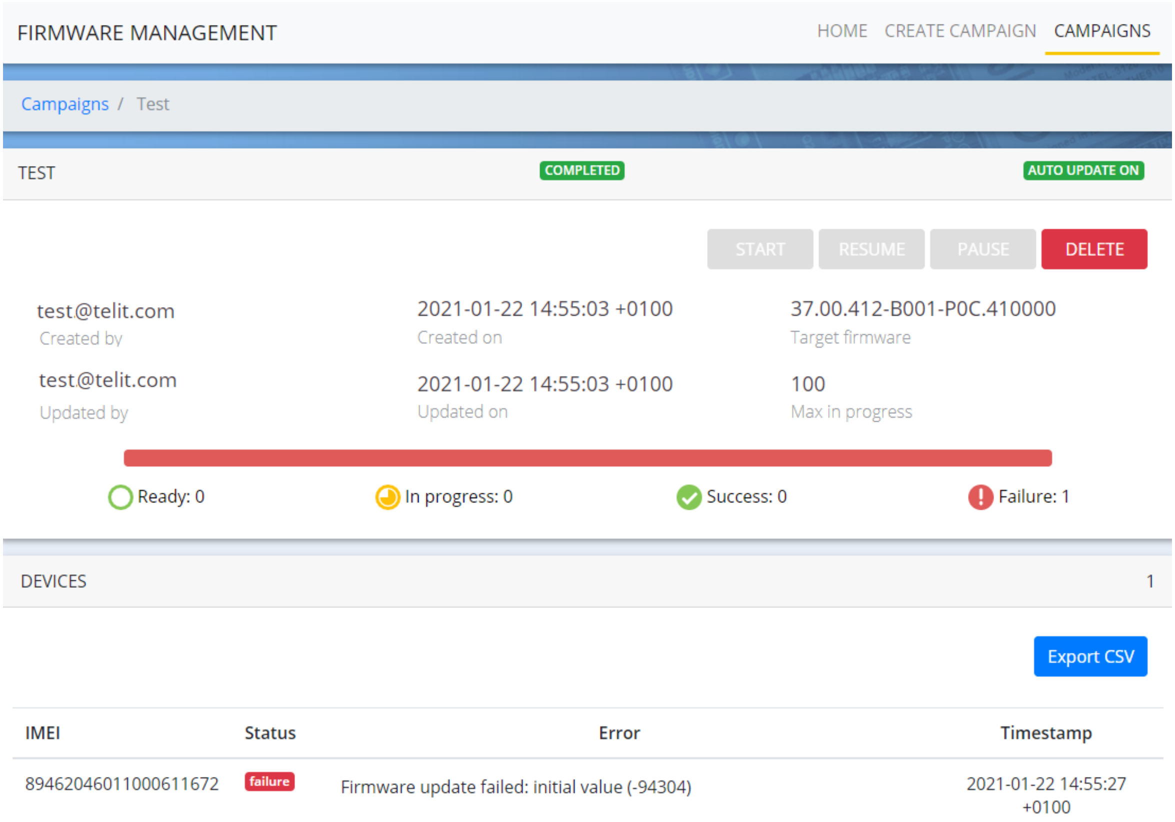


Figure 21: Firmware update failed

The User can schedule a new FOTA Campaign by starting the entire process from the beginning on the device(s) that have failed the operation.

6.2. Power Down During Delta Installation

If the electricity failure occurs during the flashing phase, the modem restarts the application of the new SW Version as soon as it is powered on again and it finishes the flashing operation. The delta image installer is designed to be failsafe and always recover from a power cycle.

The FOTA campaign remains in progress until the LwM2M client will re-register to the OneEdge portal and subsequendtly shows "Completed" status, as normal behavior.

Normally the host should wait at least 15 minutes for the URC "`#OTAEV: Module Upgraded To New Fw`", to detect that the upgrade is successfully finished (it is verified by checking the firmware version which should be "Version B").

Furthermore, the LwM2M client should automatically bootstrap and register to the portal:

```
LWM2M-TLT:"BOOTSTRAPPING",SSID=0,"coaps://bs.telit.io"
LWM2M-TLT:"BOOTSTRAPPED",SSID=0,"coaps://bs.telit.io"
```

If the above URCs are not received, the modem could be still switched off.

If the modem is powered on but the client is not automatically started, the host should consider a recovery check, according to the previous basic configurations of the modem, especially the cellular settings (i.e. AT+WS46?, apn, etc.) and restore in case of a mismatch.

Then the host can manually enable the LwM2M client, if it is disabled.

7. AT COMMAND DESCRIPTION

This section contains a detailed description of the AT commands specific for the FOTA process and also the URCs that the host can receive from the LwM2M client during the firmware upgrade process.

Note: The following commands might not be supported in specific firmware versions. Please always check the latest release of AT command guide specific to the product you are using

7.1. AT Commands

7.1.1. AT#LWM2MFOTACFG – LWM2M Client FOTA Management

This set command selects the FOTA mode that will be applied to the specified agent. The user will be able to enable the ACK request for any of the FOTA operations (delta download, delta apply for the software update, both and none of them). After the FOTA mode has been configured, an URC is issued each time the client needs the ACK to continue with the required operation. Every URC emitted should be acknowledged (and authorized) with #LWM2MFOTAACK command.

Warning: Command available from 37.00.XX1 (for MExxxG1 devices) and 30.00.XX6 (for MExxxC1 devices).
<mode> =4 available from 37.00.XX2 and 30.00.XX9 versions.

SIM Presence	Setting saved	Can be aborted	MAX timeout	SELINT
Required	Auto	No	-	2

AT#LWM2MFOTACFG=<agentInstance>,<mode>[,<timeoutAction>]

Parameters:

Name	Type	Default	Description
<agentInstance>	integer	0	selects the lwm2m instance shown below Values: 0 : Telit agent 1 : Verizon agent 2 : AT&T agent 3 : DOCOMO agent
<mode>	integer	0	select the FOTA mode Values: 0 : Normal mode, no ACK required 1 : ACK required for delta file downloading 2 : ACK required for delta application for the software upgrade 3 : ACK required for both delta download and delta update application 4 : FOTA should be rejected
<timeoutAction>	integer	0	select the action to be performed after the FOTA timeout for ACK expiration Values: 0 : After the FOTA timeout for ACK expiration, the operation waiting for user confirmation is rejected and FOTA entire operation is reset

1	:	After the FOTA timeout for ACK expiration, the operation waiting for user confirmation is automatically applied

Warning: <timeoutAction> not present and behavior as timeoutAction=1 until version 37.00.XX2 (for MExxxG1 devices) and 30.00.XX8 (for MExxxC1 devices).
timeoutAction=0 as default behavior from 37.00.XX3 (for MExxxG1 devices) and 30.00.XX9 (for MExxxC1 devices).

The FOTA timeout for ACK confirmation lasts 1 week (in case of ACK not sent) for each ACK confirmation request.

FOTA configuration <**mode**> and <**timeoutAction**> are persistent to the module power cycle.

FOTA configuration <**mode**> and <**timeoutAction**> are persistent to the factory reset and to the **#LWM2MSTS** command.

URC are emitted, according to the FOTA mode set, to allow the user to identify the operation that has triggered it. URC are issued in the format:

#LWM2MFOTARING: <agentInstance>,"DOWNLOAD"

#LWM2MFOTARING: <agentInstance>,"UPDATE"

#LWM2MFOTARING: <agentInstance>,"FOTA REJECTED"

The action that should be performed (either download or update) will be applied in a time ranging from a few seconds up to a couple of minutes.

If the <**mode**> is set to 4 (FOTA rejection), the client returns an error at every FOTA request generated by the server.

In case of FOTA operation rejected due to ACK timeout expiring without an ACK confirmation (i.e.: <**timeoutAction**> set to 0), an URC is issued:

#OTAEV: "FOTA REQUEST DROPPED"

#LWM2MFOTACFG should be given only in FOTA idle status to avoid uncertain scenarios; as soon as the delta firmware is being downloaded,every attempt to change the FOTA configuration during a FOTA operation cycle will result in an error.

AT#LWM2MFOTACFG?

Read command returns the current FOTA configuration for all the LwM2M clients currently active in the module, in the format:

#LWM2MFOTACFG: <agentInstance1>,<mode1>,<timeoutAction1>

...

#LWM2MFOTACFG: < agentInstancen>,<moden>,<timeoutActionn>

AT#LWM2MFOTACFG=?

Test command reports the range for parameters

7.1.2. AT#LWM2MFOTAACK – Ack for Telit LwM2M Agent FOTA Operation Confirmation

This command sends an ACK to the LwM2M Client to authorize the FOTA operation required to the specified client.

Warning: Command available from 37.00.XX1 (for MExxxG1 devices) and 30.00.XX6 (for MExxxC1 devices).

SIM Presence	Setting saved	Can be aborted	MAX timeout	SELINT

Required	No	No	-	2

AT#LWM2MFOTAACK=<agentIdentifier>,<action>

if the **<mode>** set with **#LWM2MFOTACFG** command is not 0 (default), the Telit LwM2M client requires an ACK to performs its operations on the dedicated data context.

Parameter:

Name	Type	Default	Description
<agentInstance>	integer	0	selects the lwm2m instance shown below Values: 0 : Telit agent 1 : Verizon agent 2 : AT&T agent 3,4 : reserved for future use
<action>	integer	1	Acknowledge to the FOTA required operation. Value: 1 : action is required

The action that should be performed (either download or update) will be applied in a time ranging from a few seconds up to a couple of minutes.

AT#LWM2MFOTAACK?

Not supported

AT#LWM2MFOTAACK=?

Test command reports the supported range of values for all the parameters.

7.1.3. AT#LWM2MFOTASTATE - LWM2M Client FOTA State

This command allows the end-user to query the module to retrieve, for a given agent, the status of the FOTA client, both in terms of the LwM2M specification status and of the internal FOTA client management status.

After the command has been issued, the returned response is in the format:

#LWM2MFOTASTATE: <fwUpdateObjStatus>,<fotaStatus>,<ackExpiringTime>

The **<fwUpdateObjStatus>** parameter returns the Firmware object resource 3 "FIRMWARE STATE", which is the FOTA status expressed in terms of the LwM2M specification.

The **<fotaStatus>** parameter returns the internal FOTA client status.

The **<ackExpiringTime>** parameter reports the time remaining until the ACK request expires, in case **#LWM2MFOTACFG** is not set to the default value, it reports the value **0** otherwise.

Warning: Command available from 37.00.XX3 (for MExxxG1 devices), not available yet for MExxxC1 devices.

SIM Presence	Setting saved	Can be aborted	MAX timeout	SELINT
Required	No	No	-	2

AT#LWM2MFOTASTATE=<agentInstance>

Parameters:

Name	Type	Default	Description
<agentInstance>	integer	0	selects the lwm2m instance shown below
	Values:		
	0 : Telit agent		
	1 : Verizon agent		
	2 : AT&T agent		
	3 : DOCOMO agent		

If the **#LWM2MFOTACFG** command is not set, the parameter **<fotaStatus>** may assume only a subset of its values set.

The **<fwUpdateObjStatus>** parameter may assume the values as in the following table:

0	IDLE
1	DOWNLOADING
2	DOWNLOADED
3	UPDATING

The **<fotaStatus>** parameter may assume the values as in the following table:

0	IDLE
1	DOWNLOAD REQUEST RECEIVED, WAITING FOR ACK
2	DOWNLOAD IN PROGRESS
3	DOWNLOAD COMPLETED
4	UPDATE REQUEST RECEIVED, WAITING FOR ACK
5	UPDATE IN PROGRESS
6	FOTA REJECTED
7	FOTA FAILED

The **<fotaStatus>** parameters values 1 (DOWNLOAD REQUEST RECEIVED, WAITING FOR ACK), 4 (UPDATE REQUEST RECEIVED, WAITING FOR ACK) and 6 (FOTA REJECTED) may be assumed only when the **#LWM2MFOTACFG** command is not set to the default value.

The **<ackExpiringTime>** parameter is expressed in seconds.

FOTA rejected is a permanent condition; this code is returned when **#LWM2MFOTACFG** is set with this option, regardless there was a FOTA request effectively.

<fwUpdateObjStatus> and **<fotaStatus>** report different intenal information, thus their value may not be constantly aligned.

In case of **#LWM2MFOTACFG** not set to the default value, once the **#LWM2MFOTAACK** is issued, the internal FOTA management may take several seconds to check it and manage it properly; during this time the **<ackExpiringTime>** parameter may continue to return decreasing values.

AT#LWM2MFOTACFG?

Read command returns:

OK

AT#LWM2MFOTACFG=?

Test command reports the range for parameters

Server URC and read command examples:

- During the Verizon agent FOTA download, **#LWM2MFOTACFG** default setting:

  AT#LWM2MFOTACFG?

  #LWM2MFOTACFG: 0,0,0

  #LWM2MFOTACFG: 1,0,0

  OK

  AT#LWM2MFOTASTATE=1

  LWM2MFOTASTATE: 1,2,0

  OK

- During the Telit agent FOTA waiting for update ACK, **#LWM2MFOTACFG** set to require ack for all the operations, after a day of waiting:

  AT#LWM2MFOTACFG?

  #LWM2MFOTACFG: 0,3,0

  OK

  AT#LWM2MFOTASTATE=0

  LWM2MFOTASTATE: 3,4,518400

  OK

- The Telit agent FOTA is set to reject all the FOTA sessions:

  AT#LWM2MFOTACFG?

  #LWM2MFOTACFG: 0,4,0

  OK

  AT#LWM2MFOTASTATE=0

  LWM2MFOTASTATE: 0,6,0

  OK

7.1.4. AT#FOTAURC - Sets FOTA Extended URCs

This command allows the end-user to enable/disable FOTA extended URCs, resulting in verbose FOTA operations. These settings are generally not related or manageable with other LwM2M agent commands.

Warning: Command available from 37.00.XX2 (for MExxxG1 devices) and 30.00.XX9 (for MExxxC1 devices).

SIM Presence	Setting saved	Can be aborted	MAX timeout	SELINT
Required	Yes	No	-	2

AT#FOTAURC=<enable>

The settings are stored in the module and are not affected by module power-cycle, TFI or FOTA flashing.

Parameter:

Name	Type	Default	Description
<enable>	integer	0	Identifier of the parameter to be set.

Values:

0 : Disable extended URCs (default)

1 : Enable extended URCs

This command affects LwM2M, FA1 and OTAUP operations.

The command should add the following URCs:

- **#OTAEV: "FOTA REQUEST INIT"** at the FOTA startup beginning
- **#OTAEV: "DOWNLOAD STARTED"** at the delta package download be
- **#OTAEV: "DOWNLOAD COMPLETED"** at the delta package downloac
- **#OTAEV: "DOWNLOAD FAILED"** at the delta package download failu
- **#OTAEV: "INTEGRITY CHECK PASS"** in case of valid delta package
- **#OTAEV: "INTEGRITY CHECK FAIL"** in case of invalid delta package

Please notice that following error URCs are issued always, regardless the Command setting:
#OTAEV: "DOWNLOAD FAILED"
#OTAEV: "INTEGRITY CHECK FAIL",<errorCode>

Please notice that the time to issue the URCs is strictly related to the agent that is in charge of managing the FOTA operations and may strongly vary between them.

Please notice that the **#OTAEV: "INTEGRITY CHECK FAIL"** URC may appear more than once, according to the retry policy of the delta validity check used by the FOTA entity, such as in LwM2M client.

<errorCode> reported in **#OTAEV: "INTEGRITY CHECK FAIL"** URC may assume the following values:

- 10 in case of invalid delta file (i.e.: when the delta file has an invalid or corrupted tag)
- 21 in case of CRC calculated error (i.e.: when the delta file is not applicable to the current software version)

AT#FOTAURC?

Returns the **<enable>** value

AT# FOTAURC=?

Test command reports the supported range of values.

7.2. URC Sequence

7.2.1. Download Message Process

URC		Description
#LWM2MFOTARING: 0,"DOWNLOAD"		Notification – wait for Delta download confirmation
Success Case	#LWM2MFOTARING: 0,"UPDATE"	Notification - Done, DELTA Firmware downloaded.
Failure Case	#OTAEV: "DOWNLOAD FAILED"	NOTIFICATION - delta package download failure

Table 1: Download Message Process

7.2.2. Update Message Process

URC		Description
#LWM2MFOTARING: 0,"UPDATE"		Notification – Delta download completed, wait for upgrade start confirmation
Success Case	#LWM2MINFO: "GEN","FOTA REBOOT" #OTAEV: Module Upgraded to New Fw	Notification - Firmware upgradE IN PROGRESS. Notification – the FOTA engine has finished its work successfully
Failure Case	#OTAEV: OTA Fw Upgrade Failed	Notification – An error occurred while the FOTA engine was doing its work

Table 2: Update Message Process

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- it can interfere with other electronic devices, particularly in environments such as hospitals, airports, aircrafts, etc.
- there is a risk of explosion such as gasoline stations, oil refineries, etc. It is the responsibility of the user to enforce the country regulation and the specific environment regulation.

Do not disassemble the product; any mark of tampering will compromise the warranty validity. We recommend following the instructions of the hardware user guides for correct wiring of the product. The product has to be supplied with a stabilized voltage source and the wiring has to be conformed to the security and fire prevention regulations. The product has to be handled with care, avoiding any contact with the pins because electrostatic discharges may damage the product itself. Same cautions have to be taken for the SIM, checking carefully the instruction for its use. Do not insert or remove the SIM when the product is in power saving mode.

The system integrator is responsible for the functioning of the final product. Therefore, the external components of the module, as well as any project or installation issue, have to be handled with care. Any interference may cause the risk of disturbing the GSM network or external devices or having an impact on the security system. Should there be any doubt, please refer to the technical documentation and the regulations in force. Every module has to be equipped with a proper antenna with specific characteristics. The antenna has to be installed carefully in order to avoid any interference with other electronic devices and has to guarantee a minimum distance from the body (20 cm). In case this requirement cannot be satisfied, the system integrator has to assess the final product against the SAR regulation.

The equipment is intended to be installed in a restricted area location.

The equipment must be supplied by an external specific limited power source in compliance with the standard EN 62368-1:2014.

The European Community provides some Directives for the electronic equipment introduced on the market. All of the relevant information is available on the European Community website:

https://ec.europa.eu/growth/sectors/electrical-engineering_en

9. GLOSSARY

ACK	Acknowledgment
APP	Application
CA	Connection Agent
DM Server	Device Management Server
DTE	Data Terminal Equipment
EVB	Evaluation Board
EVK	Evaluation Kit
FOTA	Firmware over the air
OMA	Open Mobile Alliance
RSSI	Radio Signal Strength Information
SIM	Subscriber Identification Module
TATC	Telit AT Controller
TS	Technical Support
UART	Universal Asynchronous Receiver Transmitter
URC	Unsolicited Result Code
URI	Universal Resource Identifier
USB	Universal Serial Bus

10. DOCUMENT HISTORY

Revision	Date	Changes
1	2022-06-14	Editorial changes
0	2021-03-23	First issue

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